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To: The Editor-in-Chief and Editorial Board
IEEE Transactions on Information Forensics and Security (TIFS)
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Subject: Submission of Original Research Article: "AEGIS: Non-Hermitian Quantum Walk Graph Neural Networks with Topological Point-Gap Invariants for Evasion-Resilient Network Intrusion Detection"

Dear Editor-in-Chief,

Please find enclosed our original research manuscript, **"AEGIS: Non-Hermitian Quantum Walk Graph Neural Networks with Topological Point-Gap Invariants for Evasion-Resilient Network Intrusion Detection,"** which we submit for consideration as a Regular Paper in *IEEE Transactions on Information Forensics and Security* (TIFS).

Network Intrusion Detection Systems (NIDS) are increasingly deploying spatial Graph Neural Networks (GNNs) to detect coordinated botnet infrastructures and multi-stage lateral intrusions. However, spatial message-passing GNNs harbor a fatal, provable theoretical vulnerability: bounded by the 1-Weisfeiler–Lehman (1-WL) graph isomorphism test, their multiset aggregation collapses to identical constant representations on degree-regular communication overlays when adversaries deploy flow feature mimicry. Advanced persistent threats (APTs) exploit this topological loophole by organizing lateral command-and-control (C2) overlays into degree-regular circulant topologies (k-regular graphs), successfully evading classical spatial GNNs and Markovian random walk position encodings. While closed unitary quantum walks bypass 1-WL expressivity bounds, their conservative Hermitian dynamics suffer from catastrophic zero-dissipation phase scrambling on dense networks and artificial symmetrization of asymmetric NetFlow telemetry.

To resolve these fundamental vulnerabilities, we introduce **AEGIS** (**A**symmetric **E**xceptional-point **G**raph **I**ntrusion **S**ystem), a deep open quantum walk architecture grounded in non-Hermitian operator physics, point-gap topology, and an adaptive mixture of topological and volumetric experts (AEGIS-MoE). Our manuscript provides:

1. **Theoretical Foundation & Point-Gap Symmetry Shattering (Theorems 1 & 2):** We prove that spatial GNNs and Markovian random walks experience representation collapse on degree-regular overlays under feature mimicry (Propositions 1 & 2). We establish that microscopic non-reciprocal hopping ($J_{uv} \neq J_{vu}$) induces a non-zero complex energy winding invariant $w(E_B) \in \mathbb{Z} \setminus \{0\}$, provably breaking 1-WL symmetry.
2. **Non-Hermitian Skin Effect as Frontier Condensation (Theorem 2):** Under open boundary conditions, point-gap spectral winding drives an exponential condensation of quantum probability density on directional attack boundaries, establishing a physical mechanism for localized botnet interface detection without brittle node triage heuristics.
3. **Exceptional Point Pseudospectral Sensitivity (Theorem 3):** Near order-m Exceptional Points, non-normal transient amplifications scale as $\epsilon \mapsto \epsilon^{1/m}$, sensitizing AEGIS to subtle low-and-slow C2 beaconing while environmental absorption $-i\boldsymbol{\Gamma}$ attenuates distant phase noise on dense graphs.
4. **Adaptive Mixture of Topological and Volumetric Experts (AEGIS-MoE):** We decouple continuous-time non-Hermitian transport from unnormalized directed multiset counting via a structural degree-skew router, conquering both sparse topological camouflage and dense volumetric spam floods without mutual representation degradation.
5. **Rigorous Multi-Dataset Benchmark & Out-of-Distribution Transferability:** We validate AEGIS across four standardized operational regimes (CTU-13 Scenarios 10, 9, 11, and NF-BoT-IoT). AEGIS-MoE achieves $87.11 \pm 4.60\%$ Macro-F1 on clean Scenario 10 and sustains an elevated defense floor of $69.39 \pm 4.94\%$ (and $73.74 \pm 3.02\%$ under isolated E_{topo}) under full adversarial camouflage ($\alpha = 1.00$) where classical spatial GNNs collapse toward random guessing (GIN: 53.68%, RWPE-GIN: 48.47%) and tabular GBDT

plunges to 53.89%. Furthermore, AEGIS-MoE achieves superior zero-shot cross-scenario transferability across distinct captures of the Rbot family (CTU-13 Scenario 10 $\rightarrow$ Scenario 11, $74.50 \pm 2.34\%$ Macro-F1 with 2.30% FAR at operational $\tau = 0.40$, and $72.87 \pm 3.16\%$ with 1.97% FAR and 93.02% Precision at $\tau = 0.50$) and robust multi-hop depth stability.

6. **Three-Tier Quantum Verification Architecture & Gate-Level Sz.-Nagy Dilation:** We establish an uncompromised quantum realization framework: (1) live cloud QPU execution of kinetic hopping on the 156-qubit IBM Fez transmon processor ($\bar{F} = 0.9918$); (2) gate-level physical execution of the complete non-Hermitian operator (including asymmetric hopping and dissipation) via Sz.-Nagy Unitary Dilation with ancilla post-selection under calibrated 127-qubit IBM Brisbane noise emulation ($\bar{F} = 0.962$), physically verifying the Non-Hermitian Skin Effect (36.4$\times$ boundary localization) on a gate-level quantum processor; and (3) production edge line-rate deployment (> 240 graphs/sec, < 4.2 ms latency) via classical Arnoldi Krylov projections.

7. **Systematic Dual-Expert Ablation, Tabular Baselines & Line-Rate Scalability:** Multi-seed ablation isolates E_{topo} as the sole active mechanism delivering camouflage immunity (73.74% F1 with 0.36% FAR) and E_{vol} as preventing degree-normalization starvation on volumetric floods (73.37%, $\bar{g}_{\text{topo}} = 0.74$). We benchmark against a 62-feature tabular GBDT baseline (revealing its -28.98% collapse under camouflage), verify enterprise streaming throughput exceeding 240 graphs/sec (< 4.2 ms latency), and establish the $\mathcal{O}(m^2N + m|E|)$ Krylov subspace Arnoldi projection pathway for large-scale graph monitoring.

Because *IEEE TIFS* is the premier venue for rigorous information forensics, adversarial robustness, and secure network telemetry, this work directly addresses questions of paramount importance to your readership.

This manuscript represents original work that is not under consideration for publication elsewhere. All authors have reviewed and approved the manuscript. The authors declare no competing financial or non-financial interests.

Thank you for your consideration of our work.

Sincerely,

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